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Q1 Find the value of constants a and b such that the function $f(z) = x^2 + ay^2 - 2xy + i(bx^2 - y^2 + 2xy)$ is analytic. Also find $f'(z)$

$u = x^2 + ay^2 - 2xy$

$\frac{\partial u}{\partial x} = 2x - 2y$

$\frac{\partial u}{\partial y} = 2ay - 2x$

$v = bx^2 - y^2 + 2xy$

$\frac{\partial v}{\partial x} = 2bx + 2y$

$\frac{\partial v}{\partial y} = -2y + 2x$

$u_x = v_y$

$2x - 2y = -2y + 2x$, $u_y = -v_x$

$2ay - 2x = -(2bx + 2y)$

on comparing the value of $u_y = -v_x$ we get $a = -1, b = 1$

$f'(z) = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} \Rightarrow 2x - 2y + i(2x + 2y)$

$= 2(1+i)z$

Q2 Show that the function $e^x(\cos y + i\sin y)$ is holomorphic and find its derivative

$u = e^x \cos y$

$\frac{\partial u}{\partial x} = e^x \cos y$

$\frac{\partial u}{\partial y} = -e^x \sin y$

$v = e^x \sin y$

$\frac{\partial v}{\partial x} = e^x \sin y$

$\frac{\partial v}{\partial y} = e^x \cos y$

$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$

hence C-R equations are satisfied

Therefore $f(z)$ is analytic

Now

$$f'(z) = \frac{\partial u}{\partial x} + i\frac{\partial u}{\partial x} = e^x\cos y + ie^x\sin y$$

$$= e^x(\cos y + i\sin y) = e^x \cdot e^{iy} = e^{x+iy} = e^z$$

Q3 Show that if $f(z) = u+iv$ is analytic then u and v are both harmonic function.

$z = x+iy$

$f(x+iy) = u(x,y) + iv(x,y)$

if $f(z) = u+iv$ is analytic then

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \& \quad \frac{\partial u}{\partial y} = \frac{-\partial v}{\partial x}$$

Now

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right) + \frac{\partial}{\partial y}\left(\frac{\partial u}{\partial y}\right)$$

$$= \frac{\partial}{\partial x}\left(\frac{\partial v}{\partial y}\right) + \frac{\partial}{\partial y}\left(\frac{-\partial v}{\partial x}\right) = 0$$

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = \frac{\partial}{\partial x}\left(\frac{\partial v}{\partial x}\right) + \frac{\partial}{\partial y}\left(\frac{\partial v}{\partial y}\right) = \frac{\partial}{\partial x}\left(\frac{-\partial u}{\partial x}\right) + \frac{\partial}{\partial y}\left(\frac{\partial u}{\partial x}\right)$$

$$= 0$$

Both u & v are satisfying Laplace equation

∴ Both u & v are harmonic function.

Q4 If ϕ and ψ are function of x & y satisfying Laplace equation show that $s+it$ is analytic where

$$s = \frac{\partial \phi}{\partial y} - \frac{\partial \psi}{\partial x} \quad \text{and} \quad t = \frac{\partial \phi}{\partial x} + \frac{\partial \psi}{\partial y}$$

Since ϕ and ψ are functions of x & y satisfying Laplaces eqn

$$\frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} = 0 \quad \text{and} \quad \frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} = 0$$

For the function $s+it$ to be analytic

$$\frac{\partial s}{\partial x} = \frac{\partial t}{\partial y} \rightarrow (1) \quad \text{and} \quad \frac{\partial s}{\partial s} = -\frac{\partial t}{\partial x} \rightarrow (2)$$

Now $$\frac{\partial s}{\partial x} = \frac{\partial}{\partial x}\left(\frac{\partial\phi}{\partial y} - \frac{\partial\psi}{\partial x}\right) = \frac{\partial^2\phi}{\partial x\partial y} - \frac{\partial^2\psi}{\partial x^2} \rightarrow (3)$$

$$\frac{\partial t}{\partial y} = \frac{\partial}{\partial y}\left(\frac{\partial\phi}{\partial x} + \frac{\partial\psi}{\partial y}\right) = \frac{\partial^2\phi}{\partial y\partial x} + \frac{\partial^2\psi}{\partial y^2} \rightarrow (4)$$

$$\frac{\partial s}{\partial y} = \frac{\partial}{\partial y}\left(\frac{\partial\phi}{\partial y} - \frac{\partial\psi}{\partial x}\right) = \frac{\partial^2\phi}{\partial y^2} - \frac{\partial^2\psi}{\partial y\partial x}$$

$$\frac{\partial t}{\partial x} = \frac{\partial}{\partial x}\left(\frac{\partial\phi}{\partial x} + \frac{\partial\psi}{\partial y}\right) = \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\psi}{\partial x\partial y}$$

From eq (1) (3) & (4) we get

$$\frac{\partial^2\phi}{\partial x\partial y} - \frac{\partial^2\psi}{\partial x^2} = \frac{\partial^2\phi}{\partial y\partial x} + \frac{\partial^2\psi}{\partial y^2} = \frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} = 0$$

Again eq (2), (5) and (6) we get

$$\frac{\partial^2\phi}{\partial y^2} - \frac{\partial^2\psi}{\partial y\partial x} = -\frac{\partial^2\phi}{\partial x^2} - \frac{\partial^2\psi}{\partial x\partial y}$$

$$\frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} = 0$$

Hence the function $s+it$ is analytic

Q5 verify if $f(z) = \frac{xy^2(x+iy)}{x^2+y^4}$, $z \neq 0$ is analytic at $z = 0$ or not

$z = 0$

$u + iv = \frac{xy^2(x+iy)}{x^2+y^4}$

$u = \frac{x^2y^2}{x^2+y^4}$ & $v = \frac{xy^3}{x^2+y^4}$

At the origin

$\frac{\partial u}{\partial x} = \lim_{x \to 0} \frac{u(x,0) - u(0,0)}{x} = \lim_{x \to 0} \frac{0-0}{x} = 0$

$\frac{\partial u}{\partial y} = \lim_{y \to 0} \frac{u(0,y) - u(0,0)}{y} = \lim_{y \to 0} \frac{0-0}{0} = 0$

$\frac{\partial v}{\partial x} = \lim_{x \to 0} \frac{v(0,y) - v(0,0)}{x} = \lim_{x \to 0} \frac{0-0}{x} = 0$

$\frac{\partial v}{\partial y} = \lim_{y \to 0} \frac{v(0,y) - v(0,0)}{y} = \lim_{y \to 0} \frac{0-0}{y} = 0$

$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$

Hence C-R eqn are satisfied at origin

$f'(0) = \lim_{z \to 0} \frac{f(z) - f(0)}{z} = \lim_{\substack{x \to 0 \\ y \to 0}} \left[\frac{xy^2(x+iy) - 0}{x^2+y^4}\right] \frac{1}{x+iy}$

$= \lim_{\substack{x \to 0 \\ y \to 0}} \frac{xy^2}{x^2+y^4}$

Let $z \to 0$ along the real axis $y = 0$ then

$f'(0) = 0$

along the curve $x = y^2$ then

$f'(0) = \lim_{x \to 0} \frac{x^2}{x^2+x^2} = 1/2$

Limit does not exist
Hence it is not analytic.

Q6 Show that the function defined by $f(z) = \sqrt{|xy|}$ is not regular at the origin although CR equations are satisfied.

$$f(z) = u(x,y) + iv(x,y) = \sqrt{|xy|}$$

then $u(x,y) = \sqrt{|xy|}$, $v(x,y) = 0$

At the origin $(0,0)$ we have

$$\frac{\partial u}{\partial x} = \lim_{x\to 0} \frac{u(x,0) - u(0,0)}{x} = \lim_{x\to 0} \frac{0-0}{x} = 0$$

$$\frac{\partial u}{\partial y} = \lim_{y\to 0} \frac{u(0,y) - u(0,0)}{y} = \lim_{y\to 0} \frac{0-0}{y} = 0$$

$$\frac{\partial v}{\partial x} = \lim_{x\to 0} \frac{v(x,0) - v(0,0)}{x} = \lim_{x\to 0} \frac{0-0}{x} = 0$$

$$\frac{\partial v}{\partial y} = \lim_{y\to 0} \frac{v(0,y) - v(0,0)}{y} = \lim_{y\to 0} \frac{0-0}{y} = 0$$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Hence CR equations are satisfied at origin

Now

$$f'(0) = \lim_{z\to 0} \frac{f(z) - f(0)}{z} = \lim_{z\to 0} \frac{\sqrt{|xy|} - 0}{x+iy}$$

if $z \to 0$ along the line $y = mx$ we get

$$f'(0) = \lim_{x\to 0} \frac{\sqrt{|mx^2|}}{x(1+im)} = \lim_{x\to 0} \frac{\sqrt{|m|}}{1+im}$$

Q7 Prove that the function $f(z)$ defined by

$$f(z) = \frac{x^3(1+i) - y^3(1-i)}{x^2+y^2}, \quad z \neq 0 \text{ and } f(0) = 0$$

is continuous and the C-R equations are satisfied at the origin yet $f'(0)$ does not exist.

$$f(z) = \frac{(x^3-y^3) + i(x^3+y^3)}{x^2+y^2}, \quad z \neq 0$$

$$f(z) = u + iv = \frac{x^3 - y^3}{x^2+y^2} + i\frac{x^3+y^3}{x^2+y^2}$$

$$u = \frac{x^3-y^3}{x^2+y^2}, \quad v = \frac{x^3+y^3}{x^2+y^2} \qquad z \neq 0, x \neq 0, y \neq 0$$

checking continuity u & v are rational functions of x & y $u = r(\cos^3\theta - \sin^3\theta)$ and $v = r(\cos^3\theta + \sin^3\theta)$

$x = r\cos\theta, \quad y = r\sin\theta$

$u = r(\cos^3\theta - \sin^3\theta)$ and $v = r(\cos^3\theta + \sin^3\theta)$

then $z \to 0, r \to 0$

$$\lim_{z\to 0} u = \lim_{z\to 0} r(\cos^3\theta - \sin^3\theta) = 0$$

similarly $\lim_{z\to 0} v = 0$

$$\lim_{z\to 0} f(z) = 0 = f(0)$$

$f(z)$ is continuous at $z = 0$

Hence $f(z)$ is continuous for all values of z

At the origin $(0,0)$ we have

$$\frac{\partial u}{\partial x} = \lim_{x\to 0} \frac{u(x,0) - u(0,0)}{x} = \lim_{x\to 0} \frac{x-0}{x} = 1$$

$$\frac{\partial u}{\partial y} = \lim_{y\to 0} \frac{u(0,y)-u(0,0)}{y} = \lim_{y\to 0} \frac{-y-0}{y} = -1$$

$$\frac{\partial v}{\partial x} = \lim_{x\to 0} \frac{v(x,0)-v(0,0)}{x} = \lim_{x\to 0} \frac{x-0}{x} = 1$$

$$\frac{\partial v}{\partial y} = \lim_{y\to 0} \frac{v(0,y)-v(0,0)}{y} = \lim_{y\to 0} \frac{y-0}{y} = 1$$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Hence C-R equations satisfied at the origin

Now $$f'(0) = \lim_{z\to 0} \frac{f(z)-f(0)}{z} = \lim_{z\to 0} \frac{(x^3-y^3)+i(x^3+y^3)}{(x^2+y^2)(x+iy)}$$

Let $z \to 0$ along the line $y = x$ then.

$$f'(0) = \lim_{x\to 0} \frac{0+2ix^3}{2x^3(1+i)} = \frac{i}{1+i} = \frac{i(1-i)}{2} = \frac{1+i}{2}$$

Also let $z \to 0$ along the x-axis ($y = 0$) then

$$f'(0) = \lim_{x\to 0} \frac{x^3+ix^3}{x^3} = 1+i$$

Since the limits does not exist, $f'(0)$ does not exist.

Q8 Show that the following functions are harmonic and find their harmonic conjugate.

(i) $u(x,y) = \frac{1}{2}\log(x^2+y^2)$

(ii) $$\frac{\partial u}{\partial x} = \frac{1}{2}\frac{1}{(x^2+y^2)}\cdot 2x = \frac{x}{x^2+y^2}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{(x^2+y^2)\cdot 1 - x\cdot 2x}{(x^2+y^2)} = \frac{y^2-x^2}{(x^2+y^2)}$$

$$\frac{\partial u}{\partial y} = \frac{1}{2}\cdot\frac{1}{x^2+y^2}\cdot 2y = \frac{y}{x^2+y^2}$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{(x^2+y^2)\cdot 1 - y\cdot 2y}{(x^2+y^2)^2} = \frac{x^2-y^2}{x^2+y^2}$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

Since u satisfies Laplace eq hence u is a harmonic function

$$dv = \frac{\partial v}{\partial x}dx + \frac{\partial v}{\partial y}dy$$

$$= \left(-\frac{\partial u}{\partial y}\right)dx + \left(\frac{\partial u}{\partial x}\right)dy$$

$$= \left(\frac{-y}{x^2+y^2}\right)dx + \left(\frac{x}{x^2+y^2}\right)dy$$

$$dv = \frac{x\,dy - y\,dx}{(x^2+y^2)} \Rightarrow d(\tan^{-1} y/x)$$

$$v = \tan^{-1}(y/x) + C$$

(ii) $v = \sinh x \cos y$

$$\frac{dv}{dx} = \cosh x \cos y \qquad \frac{\partial^2 v}{\partial x^2} = \sinh x \cos y$$

$$\frac{dv}{dy} = -\sinh x \sin y \qquad \frac{\partial^2 v}{\partial y^2} = -\sinh x \cos y$$

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0 \quad (v \text{ is harmonic})$$

$$du = \frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy = \frac{\partial v}{\partial y}dx - \frac{\partial v}{\partial x}dy$$

$$= -\sinh x \sin y\,dx - \cosh x \cos y\,dy \Rightarrow -[\sinh x \sin y\,dx + \cosh x \cos y\,dy]$$

$$= -d(\cosh x \sin y)$$

$u = -\cosh x \sin y + C$

Q9 In a two dimensional fluid flow the stream function is $\psi = -\frac{y}{x^2+y^2}$ find the velocity potential ϕ

$$\psi = -\frac{y}{x^2+y^2}$$

$$\frac{d\psi}{dx} = \frac{2xy}{(x^2+y^2)^2} \qquad \frac{\partial \psi}{\partial y} = \frac{y^2-x^2}{(x^2+y^2)^2}$$

$$d\phi = \frac{\partial \phi}{\partial x}dx + \frac{\partial \phi}{\partial y}dy = \frac{\partial \psi}{\partial y}dx - \frac{\partial \psi}{\partial x}dx$$

$$= \frac{y^2-x^2}{(x^2+y^2)^2}dx - \frac{2xy}{(x^2+y^2)^2}dy$$

$$= \frac{(y^2-x^2+x^2-x^2)dx - 2xy\,y\,dy}{(x^2+y^2)^2} = \frac{(x^2+y^2)dx - 2x^2dx - 2xydy}{(x^2+y^2)^2}$$

$$= \frac{(x^2+y^2)dx - x(2xdx+2ydy)}{(x^2+y^2)^2}$$

$$= \frac{(x^2+y^2)dx - xd(x^2+y^2)}{(x^2+y^2)^2} = d\left(\frac{x}{x^2+y^2}\right)$$

Integration yields $\phi = \frac{x}{x^2+y^2} + C$ where C is constant.

Q10 An electrostatic field in the xy plane is given by the potential function $\phi = 3x^2y - y^3$ find the stream function and hence find the complex potential.

Let $\psi(x,y)$ be the stream function

$$d\psi = \frac{\partial \psi}{\partial x}dx + \frac{\partial \psi}{\partial y}dy = \left(-\frac{\partial \phi}{\partial y}\right)dx + \left(\frac{\partial \phi}{\partial x}\right)dy$$

$$= -(3x^2 - 3y^2)\,dx + 6xy\,dy$$

$= -3x^2dx + (3y^2dx + 6xydy)$

on Integrating we get

$\psi = -x^3 + 3xy^2 + C$

Complex potential is given by

$w = \phi + i\psi = 3xy^2 - y^3 + i(-x^3 + 3xy^2 + c)$

$w = -i(x^3 - iy^3 + 3ix^2y - 3xy^2 - c)$

$w = -i[(x+iy)^3 + (-c)]$

$w = -iz^3 + c_1$

Q12 Find the value of the integral $\int_c \{(x+y)^2dx + x^2ydy\}$

(i) along $y = x^2$ having $(0,0)$ and $(3,9)$ as end points

(ii) along $y = 3x$ between the same points.

(i) Along $y = x^2$

$$\frac{dy}{dx} = 2x$$

$$I = \int_0^{3+9i} \{(x+x^2)dx + x^4 \cdot 2xdx\}$$

$$I = \int_0^3 (x + x^2 + 2x^5)dx = \left[\frac{x^2}{2} + \frac{x^3}{3} + \frac{x^6}{3}\right]_0^3$$

$$I = \frac{9}{2} + 9 + 243 = 256\frac{1}{2}$$

(ii) Along $y = 3x$

$$\frac{dy}{dx} = 3$$

$$I = \int_0^{3+9i} (x+3x)dx + 3x^3dx \cdot 3$$

$$I = \int_0^3 (4x + 9x^3)dx = \left[\frac{4x^2}{2} + \frac{9x^4}{4}\right]_0^3$$

$$18 + \frac{243 \times 3}{4} \Rightarrow 200\frac{1}{4}$$

Q13 Evaluate $\int_0^{1+i} (x-y+ix^2)\,dz$

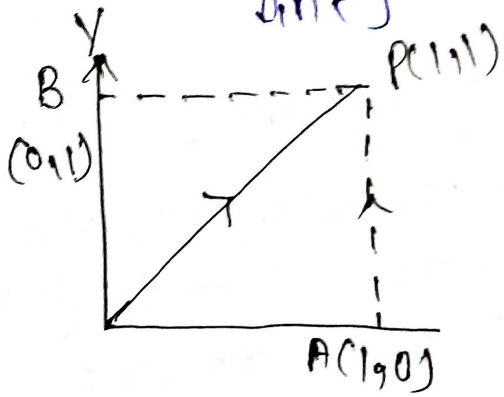
a) Along the st line from $z=0$ to $z=1+i$

b) Along the real axis from $z=0$ to $z=1$ and then along a line parallel to imaginary axis from $z=1$ to $z=1+i$

c) Along the imag. axis from $z=0$ to $z=i$ and then along a line parallel to real axis from $z=i$ to $z=1+i$

a) $\int_0^{1+i} (x-y+ix^2)\,dz = \int_0^{1+i} (x-y+ix^2)(dx+idy)$

$= \int_0^1 (x-x+ix^2)(dx+idx)$ [$y=x$ straight line]



$= \int_0^1 ix^2(1+i)\,dx = (i-1)\left(\frac{x^3}{3}\right)_0^1$

$= -\frac{1}{3} + \frac{1}{3}i$

b) $z=1$

$\int_0^{1+i} (x-y+ix^2)\,dz$

$\int_{OA} (x-y+ix^2)\,dz + \int_{AP} (x-y+iy^2)\,dz$ → (1)

⇒ Along $y=0$, $dz=dx$ $x: 0 \to 1$

$\int_{OA} (x-y+ix^2)\,dz = \int_0^1 (x+ix^2)\,dz = \left[\frac{x^2}{2} + \frac{ix^3}{3}\right]_0^1 = \frac{1}{2} + \frac{1}{3}i$

Along $x=1$ $y: 0 \to 1$

$\int_{AP} (x-y+iy^2)\,dz = \int_0^1 (1-y+i)i\,dy = \left[(-1+i)y - iy^2/2\right]_0^1$

$\Rightarrow -1+i-\frac{1}{2}i \Rightarrow -1+i/2$

Total integrate $\int_0^{1+i} (x-y+ix^2)\,dz = (1/2+1/3i)+(-1+1/2i)$

$= -1/2 + 5/6i$

(i) Along the path OBP where $z = i$

$$\int_0^{1+i} (x-y+ix^2)\,dz = \int_{OB} (x-y+ix^2)\,dz + \int_{BP} (x-y+ix^2)\,dz$$

Along $x=0$, $dz = i\,dy$, $y: 0$ to 1

$$\int_{OB} (x-y+ix^2)\,dz = \int_0^1 (-y)\,i\,dy = -i\left[y^2/2\right]_0^1 = -1/2\,i$$

Along BP $y=1$, $dz = dx$, $x: 0$ to 1

$$\int_{BP} (x-y+ix^2)\,dz = \int_0^1 (x-1+ix^2)\,dx$$

$$= \left[x^2/2 - x + \frac{ix^3}{3}\right]_0^1 = -1/2 + 1/3\,i$$

Total integral $= \int_0^{1+i} (x-y+ix^2)\,dz = -1/2\,i + (-1/2 + 1/3\,i)$

$$= -1/2 - 1/6\,i \;//$$

Q14 Evaluate $\int_0^{2+i} (\bar{z})^2\,dz$ along

(i) The real axis from $z=0$ to $z=2$ and then along a line parallel to y-axis from $z=2$ to $z=2+i$

(ii) Along the line $2y = x$

(i) ~~Along the line $2y = x$~~

Along OAP where $A(2,0)$ & $P(2,1)$

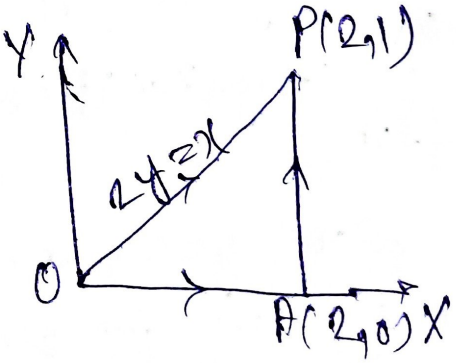
$$(\bar{z})^2 = (x-iy)^2 = (x^2-y^2) - 2ixy$$

$$\int_0^{2+i} (\bar{z})^2\,dz$$

$$\int_{OA} (x^2-y^2-2ixy)\,dz + \int_{AP} (x^2-y^2-2ixy)\,dz \quad -(i)$$

Along $y=0$ $dz = dx$ $x: 0$ to 2

$$\int_{OA} (x^2-y^2-2ixy)\,dz = \int_0^2 x^2\,dx = \left[\frac{x^3}{3}\right]_0^2 = 8/3$$



along $x=2y$, $dz=2dy$, $y: 0$ to 1

$\int_{AP}(x^2-y^2-2ixy)dz = \int_0^1 (4-y^2-4iy)2dy$

$\left[4iy - \frac{2iy^3}{3} + 2y^2\right]_0^1 \Rightarrow 4i - \frac{2}{3}i + 2 \Rightarrow 2 + \frac{10}{3}i$

Total Integral $= \int_0^{2+i} (\bar{z})^2 dz = \frac{8}{3} + 2 + \frac{11}{3}i = \frac{14}{3} + \frac{11}{3}i$

II Along $2y = x$, $dx = 2dy$

$dz = 2dy + idy = (2+i)dy$

y varies from 0 to 1

$\int_0^{2+i} (\bar{z})^2 dz = \int_0^{2+i} (x^2-y^2-2ixy)dz = \int_0^1 (4y^2-y^2-4iy^2)(2+i)dy$

$= (2+i)(3-4i)\int_0^1 y^2 dy = (10-5i)[y^3/3]_0^1 = 10/3 - \frac{5}{3}i$

Q15 Integrate $f(z) = x^2 + 3ixy$ from $A(1,1)$ to $B(2,4)$ along the curve $x = t$, $y = t^2$

$dx = dt$, $dy = 2tdt$

At $A(1,1)$ $t = 1$ and at $B(2,4)$, $t = 2$

$\int_A^B f(z)dz = \int_A^B (x^2+ixy)(dx+idy) = \int_1^2 (t^2+it^3)(dt+2itdt)$

$= \int_1^2 (t^2 - 2t^4)dt + i\int_1^2 3t^3 dt = \left[\frac{t^3}{3} - \frac{2t^5}{5}\right]_1^2 + i\left[\frac{3t^4}{4}\right]_1^2$

$= \left(\frac{8}{3} - \frac{64}{5}\right) + i\left(12 - \frac{3}{4}\right) - \left(\frac{1}{3} - \frac{2}{5}\right)$

$= \frac{-151}{5} + \frac{45}{4}i$

Q11 If $f(z) = u+iv$ is an analytic function find $f(z)$ in terms of z

~~$u = e^x(x\cos y + \cdots$~~

(i) $u - v = e^x(\cos y - \sin y)$

$if(z) = iu - v$

$f(z) + if(z) = (u-v) + i(u+v) = g(z)$

$g(z) = u + iv$

where $U = e^x(\cos y - \sin y)$

$\frac{dy}{dx} = e^x(\cos y - \sin y) = \phi_1(x,y)$

$\phi_1(z,0) = e^z(1-0) = e^z$

$\frac{dy}{dy} = e^x(-\sin y - \cos y) = \phi_2(x,y)$

$\phi_2(z,0) = -e^{-z}$

$g(z) = \int (e^z - ie^{-z})dz + C = e^z + ie^z + C = (1+i)e^z + C$

$g(z) = (1+i) f(z)$

$f(z) = \frac{g(z)}{(1+i)} = \frac{(1+i)}{(1+i)} e^z = e^z + C$

1) $u+v = \frac{x}{x^2+y^2}$

$f(z) = u+iv$

$if(z) = iu - v$

$(1+i)f(z) = (u-v) + i(u+v)$

$F(z) = U+iV$

$U = u-v, \quad V = u+v$

$V = \frac{x}{x^2+y^2}$

$\frac{\partial V}{\partial y} = \frac{(x^2+y^2)\cdot 0 - x\cdot 2y}{(x^2+y^2)^2} = \frac{-2xy}{(x^2+y^2)^2} = \phi_1(x,y)$

$\phi_1(z,0) = 0$

$\frac{\partial V}{\partial x} = \frac{(x^2+y^2)\cdot 1 - 2x^2}{(x^2+y^2)^2} = \frac{(y^2-x^2)}{(x^2+y^2)^2} = \phi_2(x,y)$

$\phi_2(z,0) = \frac{-z^2}{(z^2)^2} = \frac{-1}{z^2}$

$F(z) = \int [\phi_1(z,0) + i\,\phi_2(z,0)]\,dz + C$

$= \int 0 + (-i/z^2)\,dz + C = \frac{i}{z} + C$

$(1+i)f(z) = \frac{i}{z} + C \quad -(1)$

$f(z) = \frac{1}{(1+i)}\left[\frac{i}{z} + C\right] \quad -(2)$

$\because f(1) = 1$

from eq (1)

$(1+i)\cdot 1 = \frac{i}{1} + C$

$1+i-i = C$

$C = 1$

$f(z) =$

Putting value of c in eq (2)

$$f(3) = 2\left(\frac{1}{1+i}\right)\left(\frac{i}{2}+1\right)$$

1/(iii) $u-v = \dfrac{e^y - \cos x + \sin x}{\cos hy - \cos x}$ when $f(\pi/2) = \dfrac{3-i}{2}$

$if(z) = iu - v$

$f(z) + if(z) = (u-v) + i(u+v) = g(z)$

$g(z) = U + iV$

where $U = \dfrac{e^y - \cos x + \sin x}{\cos hy - \cos x}$

$$\frac{\partial u}{\partial x} = \frac{(\cos hy - \cos x)\cdot(0 + \sin x + \cos x) - (e^y - \cos x + \sin x)\times \sin x}{(\cos hy - \cos x)^2}$$

$= \phi_1(x, y)$

$$\phi_1(z, 0) = \frac{(1-\cos z)(\sin z + \cos z) - (1 - \cos z + \sin z)\sin z}{(1-\cos z)^2}$$

$$= \frac{\cos z - \sin^2 z - \cos^2 z}{(1-\cos z)^2} = \frac{-(1-\cos z)}{(1-\cos z)^2} = \frac{-1}{1-\cos z}$$

$$\frac{\partial U}{\partial y} = \frac{e^y(\cos hy - \cos x) - \sin hy\,(e^y - \cos x + \sin x)}{(\cos hy - \cos x)^2} = \phi_2(x, y)$$

$$\phi_2(z, 0) = \frac{1-\cos z}{(1-\cos z)^2} = \frac{1}{1-\cos z}$$

$$g(z) = \int [\phi_1 - i\phi_2]\,dz + C$$

$$= \int \left(\frac{-1}{1-\cos z} - \frac{i}{1-\cos z}\right) dz + C$$

$$= -(1+i)\int \frac{1}{1-\cos z}\,dz + C$$

$\cos 2A = 1 - 2\sin^2 A$

$2\sin^2 A = 1 - \cos 2A$

$\therefore 1 - \cos z = 2\sin^2 z/2$

$$= -(1+i)\int \frac{1}{2\sin^2 z/2}\,dz + C$$

$$g(z) = -\frac{(1+i)}{2}\left[\int \operatorname{cosec}^2 z/2\,dz + C\right]$$

$$g(z) = +(1+i)\left[\cot z/2 + C\right] \quad \text{--- (1)}$$

$$f(z) = \frac{g(z)}{(1+i)} + C = \frac{(1+i)}{(1+i)}\left[\cot z/2 + C\right] \quad \text{--- (1)}$$

given

$$f(\pi/2) = \frac{3-i}{2}$$

from eq (1)

$$\frac{3-i}{2} = (\quad)\cdot 1 + C$$

$$\frac{3-i}{2} - 1 = C$$

$$\left(\frac{1-i}{2}\right) = C$$

putting value of C in equation (1)

$$f(z) = \cot z/2 + \frac{1}{2}(1-i)$$